SEPARATED FINITE TYPE MORPHISMS OVER QCQS BASES ARE CLOSED IN SEPARATED FINITE PRESENTATION MORPHISMS

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Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Finite type is limit fp. Limits, Lemma lemma-finite-type-is-limit-finite-presentation (same chapter)
- Eventually separated. Limits, Lemma lemma-eventually-separated (same chapter)

Source reference. Original lemma 09ZQ: sources/stacks-chow/tags.txt, source lines 12608–12608. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZH: sources/stacks-chow/tags.txt, source lines 2592–2592. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1. *Let $f : X \to S$ be a morphism of schemes. Assume*

(1) *f is of finite type and separated, and*
(2) *S is quasi-compact and quasi-separated.*

Then there exists a separated morphism of finite presentation $f' : X' \to S$ and a closed immersion $X \to X'$ of schemes over S.

Proof. Apply Lemma 09ZQ and note that $X_i \to S$ is separated for large i by Lemma 01ZH as we have assumed that $X \to S$ is separated. □

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